🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Trapezoids

Lesson 5-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 2 Standard

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a trapezoid using the formula $A = \frac{1}{2}(b1 + b2) \times h$.

Language Objective: I can explain my steps using the words trapezoid, base, height, and area.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of Trapezoids | **Trapezoid** | **Parallel** | **Base 1 (b1)** | **Base 2 (b2)** | **Height** | **Perpendicular** | **Area**

👆 *Tap any word to see what it means and a picture.*

1. The space inside a trapezoid, found with $A = \frac{1}{2}(b_1 + b_2)h$, is the ______________.

2. A quadrilateral with exactly one pair of parallel sides is a ______________.

3. Two sides that stay the same distance apart and never meet are ______________.

4. One of the two parallel sides of a trapezoid is ______________.

5. The other parallel side of a trapezoid is ______________.

6. The perpendicular distance between the two parallel bases is the ________.

7. Two lines that meet at a 90° square corner are ________.

8. The amount of flat space inside the trapezoid is its ________.

Write About the Math

How many bags of concrete does the builder need?

¿Cuántas bolsas de concreto necesita el constructor?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Area of Trapezoids** — *Área de trapecios*
- ☐ **Trapezoid** — *Trapecio*
- ☐ **Parallel** — *Paralelas*
- ☐ **Base 1 (b1)** — *Base 1 (b1)*
- ☐ **Base 2 (b2)** — *Base 2 (b2)*

Model: *We use both bases because the top and bottom are different lengths.* — Students explain a trapezoid has two parallel sides of different lengths, so using only one would over- or under-count the area; strong answers mention averaging the bases.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The driveway's area is ____ sq ft because A = 1/2 x (____ + ____) x ____. He needs ____ bags because ____ / 6 = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of Trapezoids
2. **Notes 2:** Trapezoid
3. **Notes 3:** Parallel
4. **Notes 4:** Base 1 (b1)
5. **Notes 5:** Base 2 (b2)
6. **Notes 6:** Height
7. **Notes 7:** Perpendicular
8. **Notes 8:** Area
9. **Watch for this mistake:** *A common mistake in Area of Trapezoids is forgetting the ½ in $A = \frac{1}{2} \times (b1 + b2) \times h$. After adding the bases and multiplying by the height, students stop there instead of taking half — for example, with bases 6 ft and 10 ft and a height of 4 ft, they compute $(6 + 10) \times 4 = 64$ sq ft instead of taking half to get the correct 32 sq ft. That unhalved number is really the area of the parallelogram made from two trapezoids, not the area of one trapezoid. Before you submit, always ask: "Did I take half of my last answer?"*
10. **Try It 1:** A. 48 sq ft — *$A = \frac{1}{2} \times (9 + 7) \times 6 = \frac{1}{2} \times 16 \times 6 = \frac{1}{2} \times 96 = 48$ sq ft.*
11. **Try It 2:** A. 48 sq ft — *One trapezoid is half the parallelogram: $96 \div 2 = 48$ sq ft. This matches $A = \frac{1}{2} \times (5 + 11) \times 6 = \frac{1}{2} \times 16 \times 6 = 48$ sq ft — the ½ comes from the trapezoid being half of the parallelogram.*